



Koneru Lakshmaiah Education Foundation

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Recipe Sharing Platform with Meal Planner

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PROBLEM STATEMENT

Many people struggle to find suitable recipes, organize meals for the week, and maintain a structured record of their favorite dishes. Existing recipe platforms mainly focus on recipe discovery but may not provide an integrated meal-planning facility. The proposed Recipe Sharing Platform with Meal Planner provides a centralized system where users can create, share, search, save, and review recipes while also planning meals for specific days. The system helps users organize their daily and weekly meals efficiently and reduces the effort required to decide what to cook.

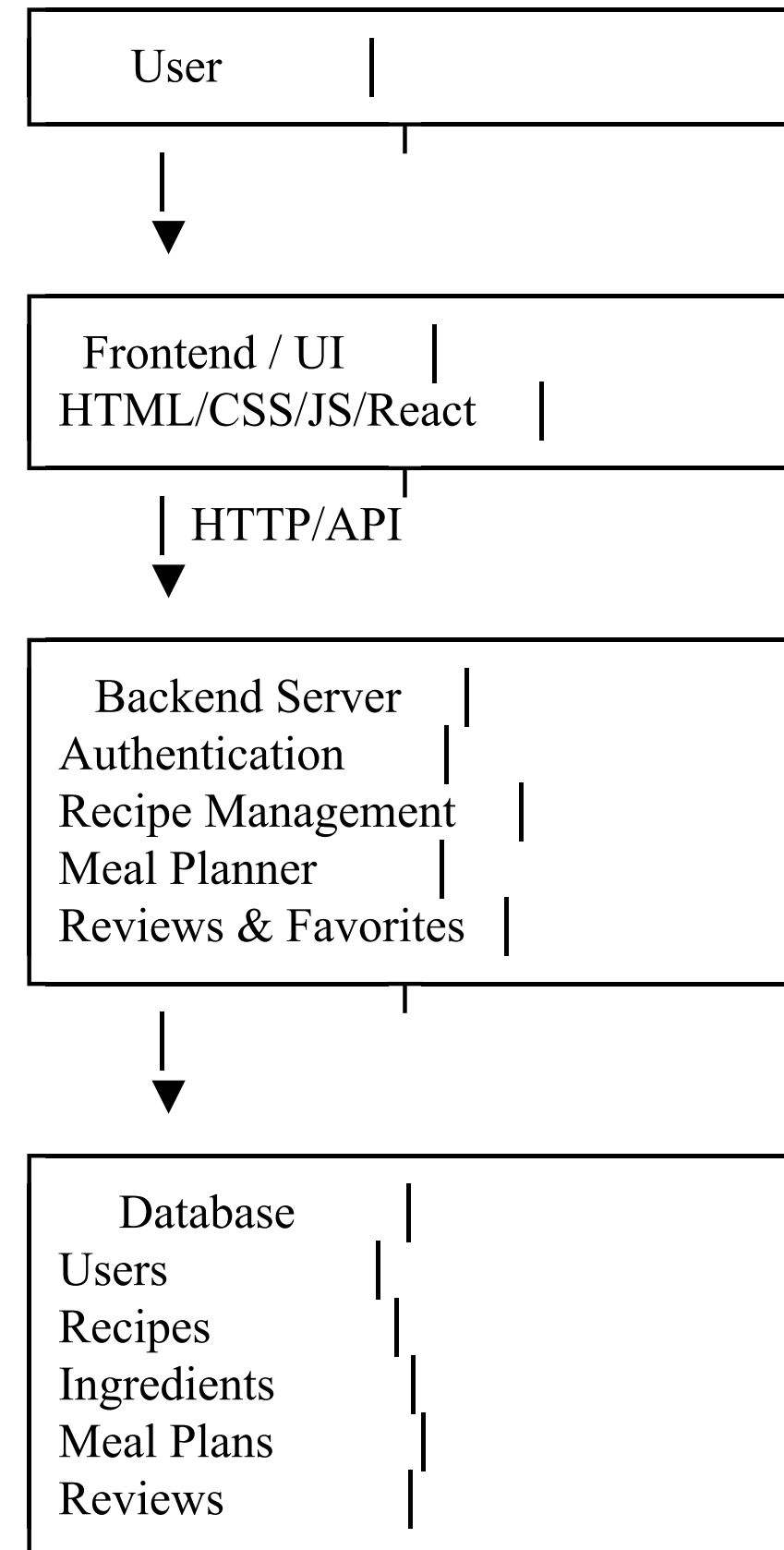
OBJECTIVES

- Develop a user-friendly platform for sharing and discovering recipes.
- Allow users to register and securely manage their profiles.
- Enable users to create, edit, delete, and share recipes.
- Provide recipe search and filtering based on categories and ingredients.
- Allow users to save favorite recipes.
- Provide a weekly meal-planning facility.
- Allow users to assign recipes to specific dates and meal types.
- Enable users to rate and review recipes.
- Maintain recipe and meal-planning information using a structured database.
- Provide an attractive, responsive, and easy-to-use interface.

PROPOSED SYSTEM

The proposed system is a web-based application consisting of a frontend, backend, and database. Users can register and log in to the platform. After authentication, they can browse recipes, search for specific dishes, view recipe details, and interact with recipes. Users can also upload their own recipes with ingredients, cooking instructions, preparation time, and images. The meal planner allows users to select a recipe and assign it to a particular date and meal type such as breakfast, lunch, dinner, or snack. Users can therefore organize their meals for an entire week. An administrator can manage users, recipes, categories, and reported content.

SYSTEM ARCHITECTURE



MODULES / FUNCTIONALITIES

User Module

- User registration
- Login/logout
- Profile management
- Password management

Recipe Module

- Add recipe
- Edit recipe
- Delete recipe
- View recipe
- Search recipes
- Filter recipes
- Upload recipe images

Ingredient Module

- Add ingredients
- Specify quantities and units
- Display ingredients for each recipe

Meal Planner Module

- Weekly/monthly calendar
- Add recipe to a date
- Select breakfast/lunch/dinner/snack
- Edit planned meals
- Delete planned meals

Favorites Module

- Add recipe to favorites
- Remove favorite recipe
- View saved recipes

Review & Rating Module

- Rate recipes
- Write reviews
- View reviews

Admin Module

- Manage users
- Manage recipes
- Manage categories
- Remove inappropriate content
- Monitor platform activity

FRONTEND DESIGN

Main Pages

1. Home Page

- Navigation bar
- Search bar
- Featured recipes
- Popular categories
- Recently added recipes

2. Login/Register Page

- Email/username
- Password
- Registration form
- Authentication buttons

3. Recipe Listing Page

- Recipe cards
- Food images
- Recipe name
- Category
- Preparation time

1. Recipe Details Page

- Recipe image
- Ingredients
- Cooking instructions
- Preparation/cooking time
- Rating and reviews
- Add to favorites
- Add to meal planner

2. Meal Planner Page

- Calendar interface
- Days of the week
- Breakfast/lunch/dinner sections
- Planned recipes

3. Profile Page

- User information
- Uploaded recipes
- Favorite recipes
- Meal plans

Figure 1 – Home Page
[Insert screenshot]

Figure 2 – User Registration/Login
[Insert screenshot]

Figure 3 – Recipe Listing
[Insert screenshot]

Figure 4 – Recipe Details
[Insert screenshot]

Figure 5 – Add Recipe Page
[Insert screenshot]

Figure 6 – Meal Planner
[Insert screenshot]

Figure 7 – User Profile
[Insert screenshot]

FRONTEND IMPLEMENTATION – SCREENSHOTS

DATABASE DESIGN

A relational database can be used to maintain structured information.

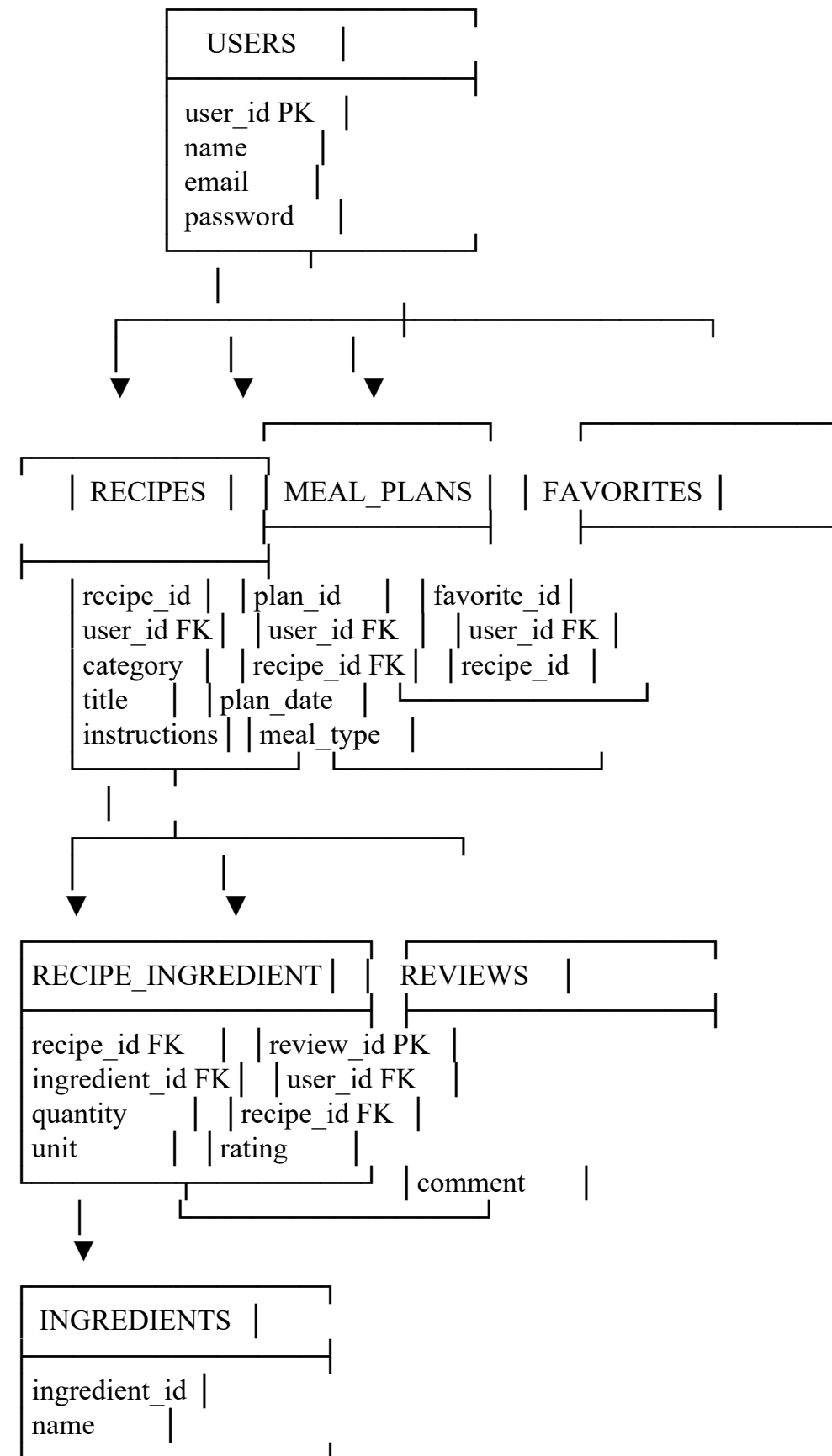
The major entities are:

- User
- Recipe
- Category
- Ingredient
- Recipe Ingredient
- Meal Plan
- Favorite
- Review

Relationships include:

- A user can create many recipes.
- A recipe belongs to a category.
- A recipe can contain multiple ingredients.
- A user can have multiple meal-plan entries.
- A user can favorite multiple recipes.
- A user can review multiple recipes.
- A recipe can receive multiple reviews.

ER DIAGRAM



DATABASE TABLES / SCHEMA

Table	Key Fields	Description
Users	user_id (PK), name, email, password_hash	Stores user registration and login details
Recipes	recipe_id (PK), user_id (FK), title, ingredients, instructions	Stores recipes shared by users
Categories	category_id (PK), category_name	Stores recipe categories such as Breakfast, Lunch, Dinner
MealPlans	plan_id (PK), user_id (FK), date, meal_type	Stores users' planned meals
Favorites	favorite_id (PK), user_id (FK), recipe_id (FK)	Stores recipes saved by users
Reviews	review_id (PK), user_id (FK), recipe_id (FK), rating, comment	Stores ratings and feedback for recipes

SAMPLE DATABASE RECORDS

Users

user_id	name	email
1	Karthikeya	karthikeya@gmail.com
2	Rahul	rahul@gmail.com

Recipes

recipe_id	user_id	title	category
101	1	Chicken Biryani	Main Course
102	2	Vegetable Pasta	Dinner
103	1	Chocolate Cake	Dessert

MealPlans

plan_id	user_id	date	meal_type	recipe_id
1	1	20/9/2026	Lunch	101
2	1	20/9/2026	Dinner	102

CURRENT PROGRESS

- Requirement analysis and project design completed.
- Database tables and relationships designed.
- User registration and login module implemented.
- Recipe creation and sharing functionality developed.
- Recipe browsing and category functionality implemented.
- Meal planner module is under development.
- Favorites and review functionality partially implemented.
- Frontend and backend integration is in progress.
- Initial testing of recipe and meal-planning features completed

CHALLENGES / ISSUES FACED

- Designing proper relationships between users, recipes, and meal plans.
- Maintaining database consistency when recipes are added or deleted.
- Implementing secure user authentication.
- Handling multiple recipes and meal plans efficiently.
- Creating an easy-to-use recipe submission form.
- Connecting the frontend with backend APIs.
- Managing validation for ingredients, instructions, dates, and other inputs.
- Making the platform responsive across different devices.

FUTURE WORK

- Add advanced recipe search and filtering.
- Add nutritional information for recipes.
- Improve the meal planner with weekly and monthly planning.
- Add automatic shopping-list generation from planned recipes.
- Add recipe recommendations based on user preferences.
- Add image uploading for recipes.
- Improve security and authentication.
- Optimize database performance.
- Add notifications and reminders for planned meals.
- Deploy the platform online for real-world users.

CONCLUSION

The Recipe Sharing Platform with Meal Planner provides a centralized platform where users can create, share, discover, save, and review recipes. The meal-planning feature helps users organize recipes according to their daily meals. The database structure maintains relationships between users, recipes, and meal plans, providing a foundation for future features such as nutrition tracking, shopping lists, recommendations, and notifications.

THANKYOU